

Joy D'Souza

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EDUCATION

Georgia Institute of Technology, Atlanta, GA

August 2024 – May 2028

Bachelor of Science in Mechanical Engineering; GPA: 4.0/4.0

Activities: Engineers Without Borders (EWB), Medical Robotics, Society of Women Engineers (SWE), ASME

WORK EXPERIENCE

SharkNinja, Product Test Engineering Intern, *Needham, MA*

May 2026 – July 2026

- Established objective KPIs for hair curl performance by translating subjective curl quality into six geometry-based metrics and developing an annotation application to measure them from tress images; validated measurement repeatability across technicians using Gage R&R and standardized application use through work instructions.
- Designed and assembled a standardized curl-retention test rig for 15 hair tresses, establishing camera alignment, lighting, and styling procedures for repeatable testing.
- Built Python tools to deterministically extract core data from DQTP and DFMEA documents, converting inconsistent formats into organized datasets that enable historical traceability for future product testing and development.

SharkNinja, Quality Engineering Intern, *Needham, MA*

May 2025 – August 2025

- Analyzed 2000+ customer reviews and troubleshooting calls to identify top product detractors and delighters across various home environment units; tracked week-over-week sentiment trends to inform product improvement priorities.
- Performed hands-on teardowns and root cause analysis (RCA) on 50+ returned units to test and identify failures with PCBAs, motors, and other internal mechanisms; documented findings, testing methods, and corrective actions.
- Conducted global cross-functional collaboration with engineering, manufacturing, product development, and customer experience teams to contribute to corrective actions with Six Sigma methods, including a rework to enhance RF signal strength between the remote and receiver on a key home environment product.

PROJECTS

Mechanical Gripper with Machine Vision, Undergraduate Research

January 2026 – Present

- Worked in a 2-person team to integrate a Raspberry Pi 5, machine vision system, and custom mechanical gripper with a UR5 robotic arm for industrial pick-and-place applications.
- Contributed 70+ labeled images to a dataset in Roboflow for training an object detection model to identify test tube location and color.
- Designed 3D-printed claw modification for existing gripper mechanism to improve object grasping performance.

Sustainable Menstrual Pad, EWB-Wish For WASH

October 2024 – Present

- Developing an adhesive-free, sustainable menstrual pad with a 14-member bioplastics team to match absorbency, comfort, and cost of industry standards (benchmarked against Always pads).
- Completed single-material absorbance testing on 9 materials per ASTM and ISO standards using water and gelatin blood simulant, identifying Himalayan nettle and wood pulp as highest mass-gain core materials.
- Progressing to composite and rewet testing to evaluate candidate materials across pad layers for absorbency, permeability, and skin comfort.

Autonomous Competition Robot, ME2110

September 2025 – November 2025

- Collaborated with 3 team members to design and fabricate a fully-autonomous robot in 8-weeks under a \$120 budget using provided mechatronics components and Arduino Uno for competition against 72 teams.
- Leveraged Onshape, Inkscape, and Bambu Studio for component fabrication on Trotec Laser machines and BambuLab P1S printers.
- Documented concept generation and evaluation, quality function deployment, and design decisions through technical reports and presentations using Morphological Charts, Engineering Specification Sheets, and House of Quality.

SKILLS

Technical: Technical Writing, RCA, 8D Problem Solving, FMEA

Software: SolidWorks, Onshape, Bambu Studio, Inkscape, Trotec JobControl, Jira, Zendesk, Microsoft Office, Minitab

Languages: MATLAB, Java, Python, HTML

Machines: Laser Cutter/Engraver, Band Saw, Miter Saw, Soldering, Instron, Mark-10

Coursework: Differential Equations, Statics, Dynamics, Deformable Bodies, Thermodynamics, Numerical Methods